

NOMINATION LETTER – KHALID A. FAKHRO, PH.D. – ARAB AWARD IN GENETICS

Dear Esteemed Award Committee Members,

Below is summary of my major achievements warranting consideration for the prestigious Arab Award in Genetics. As my contributions to this field are so deeply intertwined with the development of the field of genomics in Qatar, I offer brief commentary where necessary to provide historical context. Though unconventional, I hope this approach helps shed light on both my personal achievements in this field, as well as my unique journey to position the Arab world as a global leader in genomics.

Background Context and Commitment to the Region

In 2001, I graduated top of my country, and decided I wanted to pursue my passion for genetics – a discipline then perceived as offering limited career prospects in the GCC. After completing my BA at the University of Chicago, my Masters and PhD at Yale University (full scholarships throughout) and post-doctoral training at Weill-Cornell Medicine, I had to plan my next career move. Luckily, under the mentorship of Dr Richard Lifton, I observed first-hand the establishment of the Yale Genome Center and the utility of next generation sequencing for rare diseases. I knew how powerful this would be for my people. This ambition led to the nascent Department of Genetic Medicine at WCMC-Qatar campus.

Foundational Work and Landmark Discoveries

At the time, Qatar's local genomic research infrastructure was rudimentary. With only a small team, we spent the first few years developing foundational systems – lobbying physicians, giving departmental seminars (in both Arabic and English), identifying families, and establishing WES and computational pipelines. While the initial ramp-up was slow, by late 2016, we had sequenced >2,000 individuals, making discoveries that would spark future global interest in Arab genomics, including:

- The landmark discovery of an indigenous founder Arab population who are descendants of the earliest human migrations out of Africa (PMID: 26728717). Featured on the Feb 2016 cover of *Genome Research* (illustrated by a Qatari artist), this placed Arab genomics on the global map, catalyzing international interest in the unique genetics of our region, which was now not only attractive for high consanguinity rates, but for its remarkably under-appreciated allelic diversity.
- The first high-resolution dissection of genetic pop structure of the GCC, identifying 5 sub-populations: Gulf-, Levant-, Persian-, South Asian- and African-Qataris (24123366, 27408750).
- The first study of the impact of reference bias on genome interpretation in Arab patients, including the discovery that nearly 15% of 'variant alleles' ought to be considered 'reference' in Qataris (local allele frequency >50%), including ~200 mis-labelled as 'pathogenic' in OMIM (27408750).
- Established trio-based WES including all pipelines from scratch to support novel gene discovery, including finding patients with >1 recessive disorder due to high consanguinity (27855655).
- The creation of the first Qatari population-reference genome (QTRG), demonstrating higher alignment and variant-calling than hg19/GRCh37 for patients with Arab ancestries (27408750).
- Pioneering analysis of copy number variations in Qataris, identifying high burden for homozygous knockouts among Gulf-Arabs, and establishing tagging SNPs for ~1,000 haplotypes harboring genic deletions (26490036). Notably, same algorithms helped publish the first map of CNVs in the Tunisian population (33420067), demonstrating cross-Arab world applicability.
- The largest association study in Arabs linking >1,300 untargeted metabolites to WES data in 997 Qataris, leveraging consanguinity to fine-map 'sentinel' SNPs to missense variants (29362361).
- Identified novel alleles for diabetes and obesity (30212457, 27383215); combined metabolomics to derive metabo-clinical signatures and associated T2D comorbidities in Arabs (34732537).
- Cataloging thousands of pathogenic and likely pathogenic alleles and combining these with other biomedical and pharmacogenomic variants on the first Qatari custom-genotyping array (QChip1.0) to allow high-throughput, low-cost screening of the population (35046417).

Importantly, open-data is a core principle of my work. As we built programs and published findings, we ensured all participants consented for sample and data sharing, and contributed >1,000 Qatari exomes and genomes to the public domain – the *largest* Arab genome submission to a public database to date.

Strategy, Design and Development of the Qatar Genome Program:

In 2014, I was appointed lead consultant to the newly launched Qatar Genome Program (QGP) – the first of national sequencing program in the region and among the first worldwide. By the end of 2015, we had articulated a clear vision and mission, developed its 5-year strategy, and began recruiting the team to position Qatar as a regional genomics leader. Key initiatives we devised included:

- Establishing a centralized genomics lab at Sidra Medicine to build national sequencing capacity
- Sequencing the Qatar Biobank participants as a pilot cohort to leverage deep phenotyping data
- Establishing a strategic publication roadmap (leading to many recent high impact papers)
- Cataloguing pathogenic variants for a Qatar-specific screening microarray (Qchip 2.0),
- Launching a national grant for precision medicine research (Path towards Prec. Med., PPM)
- Increasing scientific literacy through summer internships and public engagements
- Built sustainable Arab human capital in genomics via accredited graduate programs for Genetic Counselling (at QU), and Genomics and Precision Medicine (HBKU – more details below).

Strategy, Design and Development of Human Genomics at Sidra Medicine

In 2016, following the success and high impact of our research, I was appointed inaugural Director of Human Genetics at Sidra Medicine. My goal was to build world-class genomic infrastructure and establish the best department in the region for population genomics and disease genetics. Our team's home-grown analytical pipelines formed a foundational part of advancing genome sequencing for Sidra patients, as well as analyzing large-scale WGS produced by QGP. Notable achievements:

- Established Qatar's first *point-of-care whole genome sequencing* program – the Qatari Mendelian Program; in its pilot study, we sequenced families as a first-tier test and achieved ~60% diagnostic rate (31625567). In follow-up studies, we further increased diagnostic yield using zebrafish and multi-omics (38535312, 38649831, 39937864, 34605855, 32726939).
- Established the region's largest bio-resource for Autism, the BARAKA Study, investigating the genomic and environmental factors causing ASD in Qatar (~2,000 individuals sequenced to date). Revealed significantly higher burden for recessive Autism in the local population; unexpected given the sporadic (dominant/de novo) nature of Autism worldwide (37805537, 36368308).
- Leveraged consanguinity to tackle another complex disease, male infertility; discovered and replicated 4 novel genes, significantly advancing the field for a disease that was considered largely multi-factorial with only ~10 causative genes reported at the time (29790874).
- Demonstrated the utility of long-read WGS to resolve neurological disorders, including publication of *long-read reference genomes* for 5 Arab populations; discovered >40Mb of novel sequence, suggesting we barely scratched the surface of genomics in Arab populations (40325133).
- Built a best-in-class WGS structural variant pipeline, significantly boosting diagnostic yield by identifying deleterious inversions and complex rearrangements (36329474, 37107607, 33563995, 37805537, 36368308), and discovering homozygous knockouts in QGP (*in review*)
- Resolved population structure of Qatar vis-à-vis ancient near eastern genomes, exploring Y-chromosome diversity, and constructing a *first-in-Arabs* imputation panel (34642339)
- Developed catalog of Mendelian alleles and rare variants causing extreme phenotypes in Qatar; mapped the genetic landscape of inherited treatable metabolic disorders, and included these on premarital/prenatal screening arrays (Qchip 2.0) (38584274, 35046417, 39286960)
- Conducted the first regional survey of public attitudes to genomic research (32724056), and a pioneering WISH report on Genomics and Islamic Ethics.

Notably, most studies above are firsts in any Arab population at this scale, making them foundational for Arab genomic research. Importantly, my research record demonstrates the highest commitment to quality, depth and impact, all while building infrastructure and foundational programs for the region where very little previously existed. Importantly, I invested equally in nurturing the next generation of local scientists, training students from all over the Arab world, and generating literature that is useful for all regional populations. These are vital considerations in the context of this award's requirement for a track record on the ground of significant contributions to genetics in the Arab world.

Leadership of Sidra's Research Institute: Transforming Patient Lives with Genomics

In 2019, I was appointed as Chief of Research at Sidra Medicine. I was deeply committed to transforming Sidra Medicine into a regional hub for genome-guided care. What I did not foresee was the cascade of challenges ahead: a global pandemic, serious budget pressures due to competing national priorities (World Cup), and the abrupt departure of Sidra's CEO, which led to my appointment to the Management Committee (Acting CEO). Despite these challenges, I spearheaded the most *significant growth phase* for Research at Sidra. Achievements during my tenure include:

- Enrolling thousands of patients into studies, publishing >550 papers, many in the highest tier journals worldwide (mean IF ~8.7).
- Integrating research platforms into the hospital, e.g. supporting large-scale COVID19 testing, using our microscopy core to support pathology imaging, using zebrafish lab to evaluate VUSs.
- Setting up for the first time in Qatar, *clinical-grade* exome and genome sequencing, powered by the Research Institute's Genome lab (>50,000 whole genomes – key discoveries in prev section).
- Establishing a bioibank for pediatric rare diseases, and registries for key diseases, and opening these up for local, regional and international collaborators (commitment to open-science)
- Housing Qatar's first research iPS cell lab to develop patient models and therapies
- Establishing the Sidra clinical trials office to attract early-stage trials for pediatric rare disease
- Designing and delivering Qatar's first fully accredited GMP facility supporting cell and gene therapy, and enabling, for the *first time* in Qatar, pediatric bone marrow transfer.
- Devising a visiting scientist scheme and initiating an internship program, bringing in >250 investigators/fellows/trainees/students (young Arab scientists) to our labs over the last 5 years.
- Inviting media and journalism students (Northwestern University in Qatar) to send us summer interns specializing to write stories on genetics in medicine for the public.
- Starting a national *Science Café* hosted bi-monthly at a local museum, allowing the general public to meet and engage with healthcare professionals and researchers in a casual setting.
- Hosting multiple hands-on workshops and conferences per year, including the region's longest-standing Precision Medicine and Genomics conference (11th annual PMFG will be Dec 2-4, 2025 – www.sidra.org/pmfg) where we always feature Arab genetic researchers and have travel awards for trainees to encourage their participation from afar.
- Building a research culture across the hospital, supporting clinical and translational research in the areas of rare genetic disease and genomic medicine.

Genetics Education, Training and Capacity Building: Investing in the Future of the Arab World

Alongside fostering the growth of genomics research, I have made significant contributions to genetics education in Qatar, both in terms of teaching courses as well as mentoring students on research projects, with several winning best poster/thesis awards at national/int'l conferences. I also maintained academic affiliation at both WCM-Q and WCM-NY, as well as HBKU locally. Notably, when HBKU established its College of Health and Life Sciences (CHLS) in 2016, I was part of the team that designed the graduate track in Genomics and Precision Medicine (GPM) – from setting the syllabus to teaching mendelian and population genetics, and supervising numerous students, including the *first* PhD student at HBKU and in Qatar, an Arab (Palestinian). In need of a model system to test rare variants from our Mendelian program, we used her project to establish a Zebrafish Core at Sidra Medicine. Today she manages this Core to world-class standards, and receives contracts for from around the world to model human disease genes to advance our understanding of genome biology.

Recognizing my involvement and dedication to the College, the Dean appointed me to the Graduate Program Committee (GPC), becoming its *first and only* external member to this day (now serving a 3rd term). Over the years, I have invested time and effort supporting the growth of the college and its students – supervising >20 Master's and PhD students, co-hosting conferences, teaching numerous courses, and facilitating access to Sidra Medicine for graduate students, allowing their research to have immediate impact on patients. Further, over the past six years, I developed policies at Sidra to support learning and development for all levels of Arab trainees. Watching them succeed is the single most rewarding aspect of this journey.

Thought Leadership and Global Advocacy for Arab Genomics

Over the years, I have been a vocal proponent of building Arab genomic capacity. Combined with my academic and leadership record, this earned me wide recognition as a regional thought leader, through:

- Prominent Speaker Engagements – Frequent invitations to share Qatari genetic discoveries and speak at global/regional conferences on science, precision medicine, and genomics.
- Active International Collaboration – Contributions to leading organizations (e.g., GA4GH, G2MC, ASHG), ensuring Arab perspectives are well-represented.
- Extensive Media Presence – Regular traditional and social media coverage of my work and Sidra's advancements, enhancing public genomic understanding and support for science.
- High-Level Representation – Represented Qatari genome community at the UN (Sept 2024) and at the U.S. National Academies of Science/Kuwait Foundation for Advancement of Science Precision Medicine event (October 2025).
- Building regional interest – I actively advocate for increased pharmaceutical, biotech, and venture capital investment to catalyze Middle Eastern healthcare and genomic innovation.
- Ensuing Arab representation at Sidra events – Insist on including Arab country scientists at all Sidra events and workshops, especially our flagship Precision Medicine and the Future of Genomics conference, where we invite both established and emerging Arab scientists to spotlight Arab regional programs and highlight regional progress in genomics and genetics research.

Thus, through continuous thought leadership and advocacy I have always done my best to advance genetics and genomics in the Arab world and beyond. I have said repeatedly that the next scientific renaissance will be from our part of the world, and I genuinely believe that history will prove me right.

Conclusion

In conclusion, I sometimes contemplate what might have been, had I stayed at an established university or institute in North America and decided to pursue a career in genetics research somewhere well established. I certainly chose a road less travelled, but have been lucky throughout to see my work impacting thousands of patients, inspiring hundreds of students/trainees, and setting a benchmark for genomics across the region, on par with the best ecosystems around the world.

Specifically, I believe my track record over the last decade (~90 papers and book chapters, ~\$11M in grant funding, >20 students supervised and another 250+ benefiting from processes that I put in place) demonstrates my unwavering commitment to advancing genetics in the Arab world. From catalyzing innovative genomic programs with tangible translational impact for patients, to building world-class infrastructure and teams locally, to championing education for young Arab scientists – my mission has been to transform our region from primarily knowledge consumers into significant knowledge producers, making genomic discoveries *from* the Arab world, *for* the Arab world.

The Arab Award in Genetics offers an opportunity to celebrate these accomplishments, and perhaps inspire Arab scientists to similarly follow the path less travelled (and reverse brain drain from our region). I will continue pursuing my passion and tirelessly building a future for genetics and genomics across the Arab world – doing so as a Hamdan Foundation Award recipient would, of course, be the ultimate honor.

Sincerely,



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